

AWS Modernization Platform

Cost Optimization Report

Author: Samuel Tettey-Fio

Date: June 2026

Executive Summary

This report provides an estimated monthly cost analysis for the AWS Modernization Platform and identifies optimization opportunities while maintaining security, scalability, and high availability.

Architecture Components Assessed

Amazon VPC, Transit Gateway, ALB, Auto Scaling Group, EC2, Aurora MySQL, EKS, ECR, AWS Config, GuardDuty, AWS Organizations, IAM Identity Center, CloudTrail, and S3.

Networking Layer

Transit Gateway: \$36/month.

Application Load Balancer: \$20/month.

Optimization includes removing idle resources and using VPC Peering where appropriate.

Application Layer

Two EC2 instances estimated at \$15/month.

Optimization includes Auto Scaling, Graviton instances, and stopping non-production workloads.

Database Layer

Aurora MySQL Writer and Reader estimated at \$80/month.

Optimization includes Aurora Serverless for development environments and right-sizing.

Container Platform

EKS Control Plane: \$72/month. Worker Nodes: \$30/month. Total: \$102/month.

Optimization includes Cluster Autoscaler and deleting clusters after testing.

Container Registry

Amazon ECR estimated at less than \$5/month. Optimization includes lifecycle policies and cleanup of stale images.

Security and Compliance

AWS Config: \$5/month. GuardDuty: \$10/month.

Optimization includes enabling only required rules and protections.

Cost Summary

Networking: \$56

Application: \$15

Database: \$80

Container Platform: \$102

Security: \$15

ECR: \$5

Estimated Total Monthly Cost: \$273

Recommendations

Use development environments only when needed, implement Auto Scaling, configure ECR lifecycle policies, use Aurora Serverless for non-production workloads, and destroy temporary EKS clusters after demonstrations.

Conclusion

The largest cost contributors are Amazon EKS and Aurora. Proper resource management, automation, and right-sizing can significantly reduce costs while maintaining operational excellence.